Jubair Ibn Omar Hamza – Project/Research Summary

BSC in EEE

North South University

ID: 1911226043

BS-GPA: 3.80

- ✓ **Course Name and Code:** Analog Electronics II (EEE311)

Faculty Name: Dr. Nafisa Noor (NaNr)

Project Title: IR Based Security Alarm

Project Description: The project aimed to design an IR-based security alarm system. The goal was to create a reliable and efficient alarm system using infrared technology. My role involved circuit design, component selection, and system testing. The methodology included researching IR sensors and related technologies, constructing the circuit, and conducting performance tests. The outcome was a functional IR-based security system capable of detecting movement and triggering an alert.

Hardware: Infrared LED**,** Photodiode**,** 555 Timer IC**,** Operational Amplifiers**,** Piezo Buzzer**,** Resistors, Breadboard Capacitors, and other standard circuit components

Software: Multisim, Adobe Illustrator, AutoCAD

Methodology:

a) Our methodology began with in-depth research into IR sensors and their use in security systems, setting a solid foundation for the project. We then defined our goals and established the project scope, focusing on creating an alarm system that was both highly sensitive and stable.
b) Utilized Multisim software for the virtual design, testing, and simulation of the electronic circuit. This phase involved iterative testing and optimization of the circuit for desired sensitivity and response time. Also, employed Adobe Illustrator to create clear and concise flowcharts and block diagrams. These diagrams served to visually represent the system's operation and circuit design.
c) Assembled the circuit using the listed hardware components, adhering to the design specifications. Also, conducted real-world testing to ensure the functionality and reliability of the alarm system.
d) Reviewed the project outcomes against the initial objectives, documenting the process and results.

Skills Applied:

a) Circuit design and electronic engineering
b) Analytical skills in troubleshooting and optimizing circuit performance
c) Technical skills in handling and assembling electronic components
d) Project management and collaborative work in a team environment

Achievements/Findings:

We successfully developed a compact and efficient IR-based security alarm system, showcasing the practical use of IR sensors in security applications. The project achieved an optimal balance of sensitivity and stability in the alarm's design. This endeavor also provided valuable experience in electronic circuit design, particularly in the application of infrared technology.

All necessary documents including the report paper and video clip of the project demonstration:
https://drive.google.com/drive/folders/1DQHmyNuPHywqsI3gvlWSi4Q5wzRAVYrW?usp=sharing

- ✓ **Course Name and Code:** Programming Language I (CSE115L)

Faculty Name: Mr. Rifat Ahmed Hassan (RIH)

Project Title: Secure Dynamic User Management and Shopping System

Project Description: The project is a C language-based application that demonstrates a secure and dynamic user management system coupled with a simulated e-commerce platform. The program efficiently combines various aspects of C programming, including dynamic memory management, file handling, structures, and fundamental control structures, to create a comprehensive system.

Hardware: Laptop, pc

Software/Application: Codeblocks

Programming Language: C

Skills Applied:

a) C Programming: Expertise in using C language for system-level programming.
b) Data Structures: Application of structures (struct) to create complex data types.
c) Dynamic Memory Management: Proficient use of malloc, realloc, and free for efficient memory allocation and management.
d) File I/O Operations: Skills in handling file operations for data persistence using functions like fopen, fprintf, fscanf, and fclose.
e) Error Handling: Implementing error checking and handling mechanisms to ensure the robustness of the software.
f) Problem-Solving: Applying logical and analytical thinking to develop solutions for user management and e-commerce functionalities.
g) Debugging: Identifying and resolving potential issues and bugs in the code.

Achievements/Findings:

https://drive.google.com/drive/u/6/folders/1zfAEEl-Db5UpYKIRbdzXMEkwQPTrbotb

- ✓ **Course Name and Code:** Digital Logic Design (EEE211)

Faculty Name: Dr. Mohammad Monirujjaman Khan (KMM)

Project Title: 7-Segment Sequential Circuit with 555 Timer Oscillator, Digital Counters, and Logic Gates

Project Description: The project is aimed to design and implement a sequential character display using a 7-segment common cathode display. The primary focus was to create a circuit that sequentially displays the characters B, A, C, A, in a loop, using digital electronics principles.

Methodology: In this project, I was tasked with designing and implementing a circuit to sequentially display the characters B, A, C, A on a 7-segment common cathode display, using a combination of J-K flip flops, logic gates, and a 555 timer as a clock source. My role encompassed the entire process, from conceptualization to execution, which involved virtual circuit modeling and simulation using Logisim and CircuitVerse, followed by the physical assembly and testing of the circuit.

Hardware: J-K Flip Flops, Logic Gates (ICs) - AND, NOT, OR, Switch, Capacitor, Resistor, Wiring, 9V Battery, Digital 7-Segment Display (Common Cathode), 555 Timer

Software/Application: Logisim, CircuitVerse

Skills Applied:

a) Technical Skills: Circuit design, understanding and application of digital logic, use of simulation software.
b) Analytical Skills: Problem-solving in circuit functionality, logical analysis for sequence generation.
c) Soft Skills: Attention to detail in assembling the circuit, perseverance in iterative testing and refinement

Achievements/Findings:

Successful Display Sequence: Achieved the specific sequential display (B, A, C, A) on the 7-segment display, demonstrating the practical application of digital counters and logic gates**.**

The necessary files along with a video clip of the project demonstration are here:
https://drive.google.com/drive/folders/1EOPnSh18hJnQ3ZDWEMrTEwmpjwYngOii?usp=sharing

Circuit Design and Simulation Proficiency: Gained valuable experience in using simulation tools for circuit design, which was effectively translated into a physical model.

Enhanced Understanding of Digital Electronics: The project deepened my understanding of digital electronics principles, particularly in using flip flops and timers in practical applications.

- ✓ **Course Name and Code:** Principles of Digital Communications (EEE422)

Faculty Name: Mr. Saif Ahmed (Sfa)

Project Title: Amplitude Modulation and Demodulation

Project Description: This project focused on demonstrating amplitude modulation and demodulation entirely within a MATLAB Simulink environment. My role involved designing the simulation models, executing simulations to analyze AM processes, and extracting waveform data for analysis. We utilized Simulink's comprehensive toolkit for creating and simulating the electronic circuits and signal processing blocks, allowing for detailed observation and analysis of the modulation and demodulation waveforms.

Hardware: No physical hardware was used, as the project was conducted virtually using simulation tools.

Software/Application:

A) MATLAB and Simulink for circuit design, simulation, waveform generation, and data analysis.
B) Microsoft PowerPoint for creating and presenting the project findings.

Skills Applied:

a) Technical proficiency in MATLAB and Simulink for electronic circuit simulation and signal processing.
b) Analytical skills in interpreting simulation results and waveform data.
c) Problem-solving skills in optimizing simulation models for accurate emulation of AM and demodulation
d) Public speaking and presentation skills, with a focus on clearly communicating complex technical content in English to an audience, involving the class and instructors.
e) Use of PowerPoint to effectively visualize and present project results, enhancing understanding and engagement.
f) Soft skills included teamwork, problem-solving, and effective communication

Achievements/Findings:

- ➢ Successfully simulated amplitude modulation and demodulation circuits in Simulink, demonstrating the theoretical principles and practical application of AM in telecommunications.

- ➢ Extracted and analyzed waveforms, validating the simulation models and showcasing the potential for virtual simulations in studying complex electronic systems

The project files: https://drive.google.com/drive/u/6/folders/1RXqOoQHS_AlfOaL1edo2Vj6ZsMIMSE45

- ✓ **Course Name and Code:** Junior Design (EEE299)

Faculty Name: Mrs. Meem Tasfia Zaman (MeZ)

Project Title Automatic Surface Disinfecting Device Using UV-LEDs

Project Description: The project aimed to design an Automatic Surface Disinfecting Device utilizing UV-LED technology, addressing the need for efficient and automated disinfection methods. My role involved system design, component integration, and testing. We employed a methodological approach, beginning with concept development, followed by hardware assembly and software implementation, leading to functional testing. It's a device capable of automated disinfection, demonstrating practical utility in promoting health safety. We have successfully designed a robust, effective, and efficient autonomous disinfecting (via UV-LEDS) device with the added ability to detect and avoid object, and switch off while in proximity to human motion.

Hardware:

a) UV-LED lights, Microcontroller unit (MCU) such as Arduino UNO, L293D Motor Driver shield, Motion sensors, ultrasonic sensors (HC-SR04), Power supply unit, chassis
b) Structural components for device assembly and other standard circuit components

[Note: find the detailed list in the report paper]

Software:

- Multisim for electronic circuit simulation
- Tinkercad for design, 3D modeling of the device's physical components and simulation
- Adobe Illustrator for creating presentation materials, block diagram/flow chart and schematic diagrams
- MATLAB for data analysis and algorithm development
- Arduino IDE for programming the microcontroller (Arduino UNO), using the C++ language for coding
- LTspice for additional circuit simulation and analysis

Programming Language: C++

Methodology:

This began with a comprehensive analysis of current sanitation challenges amplified by the pandemic, leading to the conceptualization of an automated disinfecting device. We utilized Arduino for control, integrating UV-LEDs for sterilization, DC motors for mobility, ultrasonic sensors for obstacle avoidance, PIR sensors for motion detection, and an L293D motor driver for motor control. Design and simulation were conducted in AutoCAD, Tinkercad, and Multisim, Proteus with Adobe Illustrator for documentation. Prototyping involved iterative development, programming in Arduino IDE using C++, and rigorous testing to ensure functionality, safety, and efficiency.

Skills Applied:

The project required a multifaceted skill set, including technical proficiency in electronics and programming, specifically in using Arduino and C++ for device control. Analytical skills were essential for designing the circuit and interpreting sensor data. Problem-solving was critical in integrating components for seamless operation. Creativity and innovation were applied in developing an affordable and efficient solution. Additionally, project

management and teamwork facilitated coordination and task allocation, while communication skills were vital for presenting the project findings effectively.

Achievements/Findings:

We have achieved our goal of creating an effective and automatic disinfection solution, demonstrating the device's capability in real-world settings. Significant findings include the device's efficiency in surface disinfection and user safety considerations during its operation.

All necessary documents including the report paper:
https://drive.google.com/drive/u/6/folders/1zgqDs3EcC6pEgLtTHSV8-RnxxvTsKGHC

- ✓ **Course Name and Code:** Senior Design (EEE499)

Faculty Name: Mr. Abu Obaidah (AbO)

Project Title: Multi-Tasking Autonomous Device

Project Description: Multi-Tasking Autonomous Device. This innovative project aims to address the heightened need for sanitation and sterilization, particularly in response to the COVID-19 pandemic, by developing a robot capable of performing various tasks such as obstacle avoidance, vacuum cleaning, fire alarm activation, UV disinfection, and light and temperature reading, biometric authentication and real time surveillance and monitoring system, all integrated into a single autonomous device. The report includes detailed sections on the background, motivation, technical design, methodology, experiment results, project impacts, planning, budget, and the engineering challenges addressed. It highlights the project's achievements, such as successful integration of advanced functionalities and improved safety and efficiency in enclosed spaces. Challenges like battery life and sensor limitations are acknowledged, with future work focusing on enhancing performance and user interaction. This project represents a significant step towards intelligent automation in living and working environments.

The features integrated into the device are as follows:

i. Obstacle Avoidance Module
ii. UV Disinfection Mechanism
iii. Fire Alarm System
iv. Light and Temperature Readings
v. Biometric Authentication
vi. Vacuum Cleaning
vii. Real time Surveillance and Monitoring System

The report paper and presentation files:
https://drive.google.com/drive/u/6/folders/1Woq_Qu3vTlZhM-vNYceyIlZkXwrpsVhy

Hardware:

a) Arduino Uno, Sonar Sensor, Servo Motor, 5V DC Motor, PIR HC-SR501, L293D Motor Driver shield, Motion sensors, ultrasonic sensors (HC-SR04), Power supply unit (batteries), double decked chassis, UV-LED lights, Photoresistor, LM35, LCD, finger print scanner etc
b) Structural components for device assembly and other standard circuit components

[Note: find the detailed list in the report paper]

Software:

- Multisim for electronic circuit simulation
- AutoCAD, Proteus and Tinkercad for design, 3D modeling of the device's physical components and simulation
- Adobe Illustrator for creating presentation materials, block diagram/flow chart and schematic diagrams
- Adafruit software libraries for fingerprint scanning.
- Arduino IDE for programming the microcontroller (Arduino UNO), using the C++ language for coding
- MATLAB for data analysis and algorithm development
- LTspice for additional circuit simulation and analysis

Programming Language: C++ (for Arduino and Tinkercad)

Skills Applied:

- Technical Engineering: Design and assembly of electronic circuits, integration of sensors, and actuator systems
- Software Development: Writing and debugging code for microcontroller programming, incorporating libraries for specific modules like the Adafruit fingerprint scanner.
- Analytical Problem-Solving: Systematically troubleshooting issues during the integration phase and optimizing performance.
- Project Management: Coordinating the project's multiple facets to ensure timely and cohesive progress.
- Innovation: Creatively combining various technologies to produce a novel autonomous system.
- Collaboration: Working effectively in a team, leveraging the diverse expertise of each member.
- Communication: Presenting complex technical information in a clear and understandable manner.

Achievements/Findings:

- Development of a Multi-Tasking Autonomous Device: The successful development of a versatile device capable of performing multiple tasks autonomously

- Integration of Advanced Functionalities: The integration of multifaceted modules such as UV disinfection, biometric authentication, environmental sensors, and surveillance capabilities into a single device, providing users with a comprehensive and efficient solution.

- Efficient Cleaning and Disinfection: The device's automated cleaning mechanism, coupled with UV disinfection, ensures thorough and effective cleaning, reducing the manual effort required and providing a hygienic environment

- ❖ Enhanced Safety and Security: The inclusion of a fire alarm mechanism and biometric authentication adds an extra layer of safety and security, alerting occupants in case of emergencies and allowing secure access control.

- ❖ Real-Time Environmental Monitoring: The integration of light and temperature sensors enables real-time monitoring of the surrounding environment, providing valuable data for users to maintain optimal conditions.

- ❖ Remote Surveillance and Monitoring: The inclusion of a Wi-Fi mini camera enables users to remotely monitor their spaces, enhancing security and allowing for convenient surveillance from connected devices.

- ❖ User-Friendly Design: The device is designed with user convenience in mind, ensuring ease of operation, intuitive controls, and a compact form factor that fits seamlessly into various indoor environments.

The report paper and presentation files:
https://drive.google.com/drive/u/6/folders/1Woq_Qu3vTlZhM-vNYceyIlZkXwrpsVhy

- ✓ **Course Name and Code:** Direct Research (EEE498R)

Faculty Name: Dr. Dihan Md. Nuruddin Hasan (DMH)

Project Title: Investigating the Capabilities of Multi-tasking Autonomous Devices: A Comprehensive Study and Implementation

Description:

This paper presents an in-depth exploration into the development and application of autonomous devices capable of performing multiple tasks simultaneously. This research, forming the crux of my Senior Design Project (EEE499A & 499B), delves into the integration of advanced robotics, AI algorithms, and sensor technologies to create a device that autonomously navigates and executes tasks in dynamic environments. The project encompasses designing, prototyping, and testing phases, highlighting challenges, solutions, and advancements in autonomous systems. Through rigorous analysis and experimentation, we contribute novel insights and methodologies to the field of robotics, emphasizing efficiency, adaptability, and scalability of multi-tasking capabilities in autonomous devices.

Report Paper: https://drive.google.com/drive/u/6/folders/1-f9njWbK-GacgY3IrnB2nhGVwiCsD3sK

Jubair Ibn Omar Hamza – Project/Research Summary

Post Graduate Diploma (not completed yet)

United International University (UIU)

ID: 905233020

- ✓ **Course Name and Code:** PGDDS 101 (A) : Programming on Core Python

Course Facilitator: Mr. MD. Aksadur Rahman

Project Title: Ecommerce Purchases Overview

Project Description: The project aimed to analyze ecommerce purchase data to understand customer behavior and trends. My role involved data analysis, focusing on extracting insights from customer purchase patterns using pandas in Python. The methodology involved data cleaning, exploratory data analysis, and statistical analysis to identify key metrics like average spend, popular items, and customer demographics. Outcomes included actionable insights on customer preferences and spending habits.

Hardware Used: Standard computing device with sufficient processing power for data analysis (i.e. PC, Laptop)

Software/Tools Used: Python, Jupyter Notebook, pandas for data manipulation, matplotlib and seaborn for data visualization.

Programming Language: Python

Skills Applied: Data analysis, Python programming, statistical analysis, data visualization, critical thinking, problem-solving.

Achievements/Findings:

Identified trends in customer spending, most popular items, demographic insights, and recommendations for targeted marketing strategies. Significant findings included the identification of high-value customer segments and opportunities for increasing sales through targeted promotions.

All necessary documents:

https://drive.google.com/drive/u/6/folders/1YO6IHqPclek_pBMctaqQoATv5WlOAfiX

- ✓ **Course Name and Code:** PGDDS 102 (A): Machine Learning using Python

Course Facilitator: Mr. MD. Aksadur Rahman

Project Title: Laptop Price Predictor

Project Description:

The project focused on developing a machine learning model to predict laptop prices based on various features such as brand, processor type, RAM, and more. My role encompassed data preprocessing, feature engineering, model selection, and evaluation. We employed linear regression and random forest models, evaluated them using cross-validation, and optimized for accuracy.

Hardware Used: A standard personal computer.

Software/Tools Used: Pycharm, python, Jupyter Notebook, numpy, pandas, scikit-learn, matplotlib, sklearn and seaborn.

Programming Language: Python

Skills Applied:

Data cleaning, linear regression, encoding, exploratory data analysis, machine learning, model evaluation, programming in Python.

Achievements/Findings:

Successfully developed a model that predicts laptop prices with high accuracy, providing insights into key factors affecting laptop prices (accuracy > 80%).

All necessary documents:
https://drive.google.com/drive/u/6/folders/1pNnhFDx2VWNHuAT81lEfrTfqJlKo8aDs

- ✓ **Course Name and Code:** PGDDS 103 (A): Data Analysis & Data Visualizations

Course Facilitator: Mr. Mohammad Ali

Project Title: Predictive Analytics in the Automotive Industry: A Machine Learning Approach to Car Price Estimation

Project Description:

The project described in the document involves the prediction of car prices using various features. The methodology includes data preprocessing, exploratory data analysis, and model training with Decision Tree Regressors and Linear Regression. The project utilized Python libraries such as pandas, seaborn, scikit-learn, and matplotlib for data manipulation, visualization, and modeling.

Hardware: Laptop, PC etc

Software/Programming language/library: Jupyter Notebook, Python, pandas, seaborn, scikit-learn, matplotlib, sklearn, numpy, scipy etc

Skills Applied:

Data cleaning and preprocessing, feature engineering, logistical regression, exploratory data analysis, statistical analysis, machine learning modeling, evaluation and data visualization.

Achievements/Findings:

I successfully developed predictive models with significant accuracy in estimating car prices, utilizing key features like engine size, curb weight, and others. The findings included insights on feature importance and model performance metrics, demonstrating the feasibility of predicting car prices with available data.

All necessary documents:
https://drive.google.com/drive/u/6/folders/1gxSDwcAA2XpdRdm06hK09LEHsU9pb_-M